

Probabilistic Choice and Quantal Response Equilibrium

John Van Huyck

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Outline

- ▶ Luce (1959) on Algebraic versus Probabilistic Choice Theory
- ▶ Theory versus Econometrics

Review of Discrete (Quantal) Choice Models

- ▶ Consider a person who must choose an alternative, a , from a set of mutually exclusive alternatives, A .
- ▶ Neoclassical economist's approach is to assume a utility function
 - ▶ ranks alternatives in a consistent and unambiguous manner
 - ▶ Person chooses alternative that is ranked first
 - ▶ The choice process is deterministic
- ▶ Psychologists, at least since Fechner (1860), view choices as the outcome of a probabilistic process.
 - ▶ human behavior appears inconsistent
 - ▶ a person making repeated choices between two alternatives under similar circumstances sometimes chooses one and sometimes the other
 - ▶ possible to find situations in which people appear to make intransitive choices, see Mays (1956)
 - ▶ Such behavior reflects fluctuations that are inherent in the process of evaluating alternatives.

Deterministic versus Probabilistic Choice

- ▶ Observed inconsistencies arise because people are sensitive to different aspects of the alternatives under different choice situations, see Stigler and Becker (1977).
- ▶ If we could identify the reasons that lead a person to favor some aspects over others, then we could work in a deterministic framework.
- ▶ However, such a model would have too many contingencies to be tractable.
- ▶ Preferable to use probabilistic choice models
- ▶ Such models can yield qualitatively different results than deterministic choice models.

Luce's (1959) axiomatic approach to Probabilistic Choice

- ▶ Luce (1959) argued that choice behavior is best described as a probabilistic, not an algebraic, phenomenon.
 - ▶ When a person reaches a decision between a and b there is a probability $P(a, b)$ that the choice will be a rather than b .
 - ▶ These probabilities will generally be different from 0 and 1.
 - ▶ The alternative is to suppose that the probabilities are always 0 and 1 and that the observed choices tell us which it is;
 - ▶ In this case, the algebraic theory of relations seems to be most appropriate mathematical tool.
- ▶ He doubts that the choice between the two approaches can be settled empirically
- ▶ The choice will ultimately depend on which theory gives a more parsimonious and complete explanation of the total range of phenomena

The Luce Model

- ▶ The choice axiom is a relation between choice probabilities defined on different subsets of A .
- ▶ Denote the probability that the individual chooses $a \in S$ when confronted with choice set S by $P_S(a)$.
- ▶ Choice axiom: for any $S \subseteq A$ and $T \subseteq A$ such that $S \subseteq T$
 - ▶ (i) if, for given $a \in S$, $P(a, b) \neq 0, 1$ for all $b \in T$, then $P_T(a) = P_T(S) \times P_S(a)$;
 - ▶ (ii) if $P(a, b) = 0$ for some $a, b \in T$, then for all $S \subseteq T$ we have $P_T(S) = P_{T-\{a\}}(S - \{a\})$.

Theorem (Luce 1959, p. 23)

Theorem

Assume that $P(a, b) \neq 0, 1$ for all $a, b \in A$. Part (i) of the choice axiom is satisfied if and only if there exists a positive real-valued function u defined on A such that

$$P_s(a) = \frac{u(a)}{\sum_{b \in S} u(b)}.$$

Furthermore, this function is unique up to multiplication by a positive constant.

Interpretation

- ▶ The function $u(a)$ can be interpreted as a deterministic “utility” function of the alternative a .
- ▶ The probability that a is chosen is increasing in its utility and is decreasing in the utility of the other alternatives.
- ▶ Using the transformation $\mu(a) = \ln u(a)$,
- ▶ we can write $P_s(a)$ as

$$P_s(a) = \frac{\exp \mu(a)}{\sum_{b \in S} \exp \mu(b)}$$

- ▶ which is the multinomial logit model. That is, the choice axiom and the multinomial logit model are equivalent.
- ▶ For the binomial case we can write

$$P(a, b) = \frac{1}{1 + \exp(\mu(b) - \mu(a))}$$

Models with Stochastic Utility

- ▶ Thurstone (1927, p. 380) “there is some qualitative fluctuation [in responses] from one occasion to the next ... for a given stimulus.”
- ▶ For example, People don't always pick the “louder” tone, the one with more sonic energy, especially when they are close together.
- ▶ Thurstone model
 - ▶ (1) a given stimulus provokes a “sensation” that is the realization of a random variable
 - ▶ (2) the response of an individual having to compare two stimuli results in the comparison of two random variables that represent the sensations provoked by the stimuli.
 - ▶ index the alternatives in A by $i \in I = \{1, \dots, n\}$
 - ▶ let $e_1, \dots, e_n$ be random variables

$$P_A(i) = Pr \left[u_i + e_i = \max_{j \in I} (u_j + e_j) \right], \quad i \in I$$

Double Exponential Distribution and The Multinomial Logit

Theorem

Assume that the e_i are i.i.d. according to the double exponential distribution

$$F(x) = \Pr(e_i \leq x) = \exp\left[-\exp\left(-\frac{x}{\mu} + \gamma\right)\right],$$

where γ is Euler's constant (≈ 0.5772) and μ is a positive constant. Then the resulting choice probabilities are given by

$$P_A(i) = \frac{\exp\left(\frac{u_i}{\mu}\right)}{\sum_{j \in I} \exp\left(\frac{u_j}{\mu}\right)}, \quad i \in I.$$

Which is the multinomial logit model.

Notation for Strategic Form Game

- ▶ A strategic form game G consists of
 - ▶ An index set of players: $i \in I = \{1, 2, \dots, n\}$
 - ▶ A pure strategy space for each player: S_i for all $i \in I$.
 $S = \times_{i \in I} S_i$
 - ▶ The observed payoff functions $u_i(s)$ for each strategy combination $s = (s_1, \dots, s_n)$ for all $i \in I$.

Latent variable derivation of logit response function

- ▶ The latent payoff functions $u_i(s) + v_i(s_i)$ for all $i \in I$,
 - ▶ where $v_i(s_i)$ is an absolutely continuous random vector on a finite interval
 - ▶ The random payoff shock $v_i(s_i)$ is private information to each player.

- ▶ Let σ denote a mixed strategy combination:

$$\sigma \in \times_{i \in I} \Delta(S_i) = \Delta$$

- ▶ Let player i 's response distribution $BR_i(\sigma_{-i})$ be given by

$$BR_i(\sigma_{-i})(s_i) = \Pr[v_i \text{ such that } s_i \text{ is a best response to } \sigma_{-i}]$$

- ▶ If $v_i(s_i)$ has a double exponential cumulative density function, then we get the logistic response function:

$$BR_i(\sigma_{-i})(s_i) = \frac{\exp^{\lambda u_i(s_i, \sigma_{-i})}}{\sum_{r_i \in S_i} \exp^{\lambda u_i(r_i, \sigma_{-i})}}$$

- ▶ where $\lambda = \frac{1}{\mu}$

Logit Equilibrium in Finite Strategic Form Games

- ▶ A logit equilibrium is a fixed point of the logit response functions, that is, a strategy combination σ^* is a *Logit Equilibrium* if $BR_i(\sigma_{-i}^*) = \sigma_i^*$ for all $i \in I$.

Theorem

Every finite strategic form game has a logit equilibrium.

Proof.

The logit response functions define a continuous function from Δ into Δ , where Δ is the product space of n simplexes, and, hence, by Brouwer's fixed point theorem it has a fixed point. □

Nash Equilibrium is a special case of Logit Equilibrium

Definition

The *logit equilibrium correspondence* is given by

$$\sigma^*(\lambda) = \left\{ \sigma \in \Delta \mid BR_i(\sigma_{-i})(s_i) = \frac{\exp^{\lambda u_i(s_i, \sigma_{-i})}}{\sum_{r_i \in S_i} \exp^{\lambda u_i(r_i, \sigma_{-i})}} \quad \forall s_i \in S_i \forall i, -i \right\}$$

Theorem

Let $\{\lambda_1, \lambda_2, \dots\}$ be a sequence such that $\lim_{t \rightarrow \infty} \lambda_t = \infty$ and let $\{p_1, p_2, \dots\}$ be a corresponding sequence with $p_t \in \sigma^*(\lambda_t)$ for all t such that $\lim_{t \rightarrow \infty} p_t = p^*$, then p^* is a Nash Equilibrium.

Proof.

Proof is by contradiction. Assume p^* assigns positive probability to an inferior response and then show that in the limit the logistic response function assigns zero probability to the inferior response.



Using Logit Equilibrium to select amongst Nash Equilibria

Theorem

For almost all finite strategic form games

1. the cardinality of $\sigma^*(\lambda)$ is odd for almost all λ ;
2. The graph of σ^* contains a unique branch which starts at the centroid, for $\lambda = 0$, and converges to a unique Nash Equilibrium, as λ goes to infinity.

Discussion

- ▶ The Logit Equilibrium integrates important results from psychology, game theory, econometrics, and the science of strategic behavior.
- ▶ The Logit Response Function has nice properties
 - ▶ It can be derived axiomatically
 - ▶ It can be derived from a latent variable model of rational choice
 - ▶ It is a function rather than a correspondence
 - ▶ So one can use Brouwer's fixed point theorem to prove the existence of logit equilibria;
 - ▶ Learning models converge to Logit equilibria, because if a player's assessment converges, then his behavior will also.
 - ▶ It solves the problem of observing zero probability events formally.
 - ▶ It can be estimated using standard maximum likelihood techniques.

Discussion continued

- ▶ The Logit equilibrium is useful from the view point of both theory and science:
 - ▶ Nash equilibria including mixed strategy equilibria can be obtained as the limit of logit equilibria
 - ▶ The graph of the logit equilibrium correspondence can be used as an equilibrium selection theory
 - ▶ Logit equilibria make more accurate predictions about human behavior in laboratory experiments

Literature

- ▶ Psychologists like Thurstone (1927) and Luce (1959) believe that human behavior is stochastic not deterministic
 - ▶ Luce references experiments that test the choice axiom directly and experiments that test the derived multinomial logit model.
 - ▶ Thurstone's law of comparative judgement
 - ▶ The probability people can discriminate between two sound energies increases as the difference in sound energy increases
 - ▶ Even with very large differences in sound energy people's ability to discriminate is not perfect.
- ▶ Some economists, like Chen, Friedman, and Thisse (1997), argue that people's ability to discriminate between two sources of utility is similarly imperfect.

Literature continued

- ▶ Fudenberg and Kreps (1993), McKelvey and Palfrey (1995), and Fudenberg and Levine (1998) use modifications of Harsanyi's (1973) purification theorem to construct what Fudenberg and Levine call "Nash distributions" and McKelvey and Palfrey call "Quantal Response Equilibria" QRE.
- ▶ Both give the logit equilibrium as a closed form example.
- ▶ Haile, Hortacsu, and Kosenok (2008) show
 - ▶ that QRE imposes no falsifiable restrictions
 - ▶ and the Logit Equilibrium, which does impose falsifiable restrictions, may impose undesirable restrictions
- ▶ Goeree, Holt, and Palfrey (2005) suggest a axiomatic derivation of regular QRE, which have falsifiable restrictions.